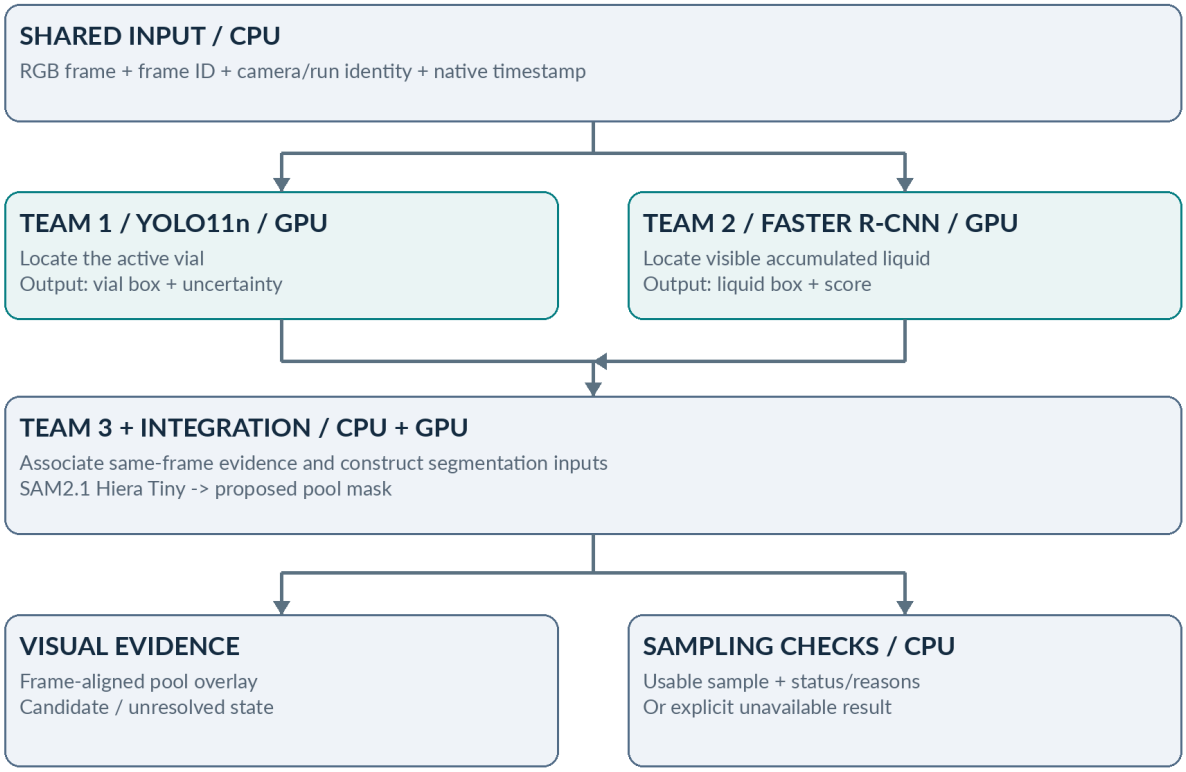
TEAM IMPLEMENTATION BRIEF

Build the liquid vision pipeline

A shared architecture, three teams, independently justified designs

Goal: detect an accumulated liquid pool as early as it becomes visibly identifiable in the active vial, produce a useful mask, and report whether the image supports a reliable sample. Each team owns a module and must explain the evidence and assumptions behind its decisions.



Same frame in both detector branches. Review labels are evaluation evidence, not inference inputs.

Starting model families

Team	Starting point	Primary responsibility
1	YOLO11n	Identify the active filling-station vial and provide dependable geometry.
2	Faster R-CNN / ResNet50-FPN	Locate accumulated liquid, especially shallow early pools.
3	SAM2.1 / Hiera Tiny	Design mask prompting/refinement and integrate frame-aligned outputs.

The detector branches receive the same image in parallel. CPU work includes decoding, association, sequence state, checks and presentation; learned detector and SAM inference are GPU stages. This handout defines responsibilities and contracts. Teams choose and justify the implementation details.

SHARED INTERFACES

Make the modules connect

Agree on data meaning before comparing model outputs

Interface	Required information
Frame input	Original RGB, camera/run identity, unique sample/frame ID, width/height and native video timestamp when available. Source video is 640 × 480 at nominal 30 fps; choose and document the analysis cadence.
Vial output	A proposed active-vial box, confidence/uncertainty and matching source identity. If the active vial cannot be identified reliably, report that state explicitly.
Liquid output	Proposed liquid region(s), scores and matching frame identity. Accumulated pool, falling stream and glass are distinct evaluation concepts.
Mask output	A full-image binary pool-mask proposal with its source identity and prompt provenance. A transport PNG uses 0 outside and 255 inside; no resized display coordinates.
Sampling output	Usable/unavailable state with reasons; an exact sampling region and image-derived feature only when its supporting checks pass.
Integration	One camera view with frame-aligned overlays and clear unresolved/unavailable states. Keep visualization, detection and sampling outcomes distinguishable.

Coordinate contract

The shared module interface uses full-image pixel [left, top, right, bottom] coordinates, with an exclusive right/bottom bound. YOLO training files use normalized center/width/height; COCO files use pixel left/top/width/height. The data guide specifies those conversions. Model-specific resized coordinates must be mapped back before another team consumes them.

Evidence boundaries

A mask proposal does not by itself establish a valid measurement or product-quality decision. A missing proposal is not automatically proof of an empty vial. Camera station IDs are not verified physical vial or recipe IDs. Keep review truth separate from inference inputs.

Still capture timestamps are not verified video-frame positions. Use the supplied source index, decoded frame number and native PTS where present; never invent a still/video alignment from an assumed frame rate.

DESIGN & EVALUATION

Questions your team must answer

Implement, test and explain your own approach

Team 1 — vial geometry

How will you distinguish the active vial from neighbors? What should happen during carrier movement, occlusion or a vial change? How will incorrect geometry affect the other modules, and how will you expose uncertainty?

Team 2 — earliest visible pool

What visual evidence distinguishes accumulated liquid from a falling stream, reflections or wet glass? Which examples are truly empty versus uncertain? How will you evaluate shallow-pool detection without using the model's own predictions as truth?

Team 3 — mask refinement and integration

What information should prompt the segmenter? How will your design behave when detector boxes are oversized, inaccurate or absent? How should sequence information help without borrowing evidence from the future or another vial? What makes a sample usable?

Required demonstrations

Deliverable	What to show
Data understanding	Explain class semantics, coordinate conventions, source splits and which labels remain provisional.
Module evidence	Evaluate boxes, masks or sequence behavior using suitable reviewed references. Report failures and uncertainty as well as successful examples.
Integrated recording	Show one frame-aligned output view covering early fill, established fill and a carrier/vial transition. Keep sample availability separate from pool presence.
Method explanation	Explain the role of each model, the decisions surrounding it and why your chosen evidence supports those decisions.
Evaluation plan	Specify how you would measure onset delay, false alarms, mask quality, temporal consistency and end-to-end runtime.

The starter data supplies RGB and provisional boxes, not an answer key for masks, timing or final prompting/state rules. Existing runs have been inspected; use a new source-disjoint recording for a blind final test. Public background: Ultralytics YOLO11 documentation; TorchVision Faster R-CNN ResNet50-FPN documentation; Meta SAM 2 model reference.